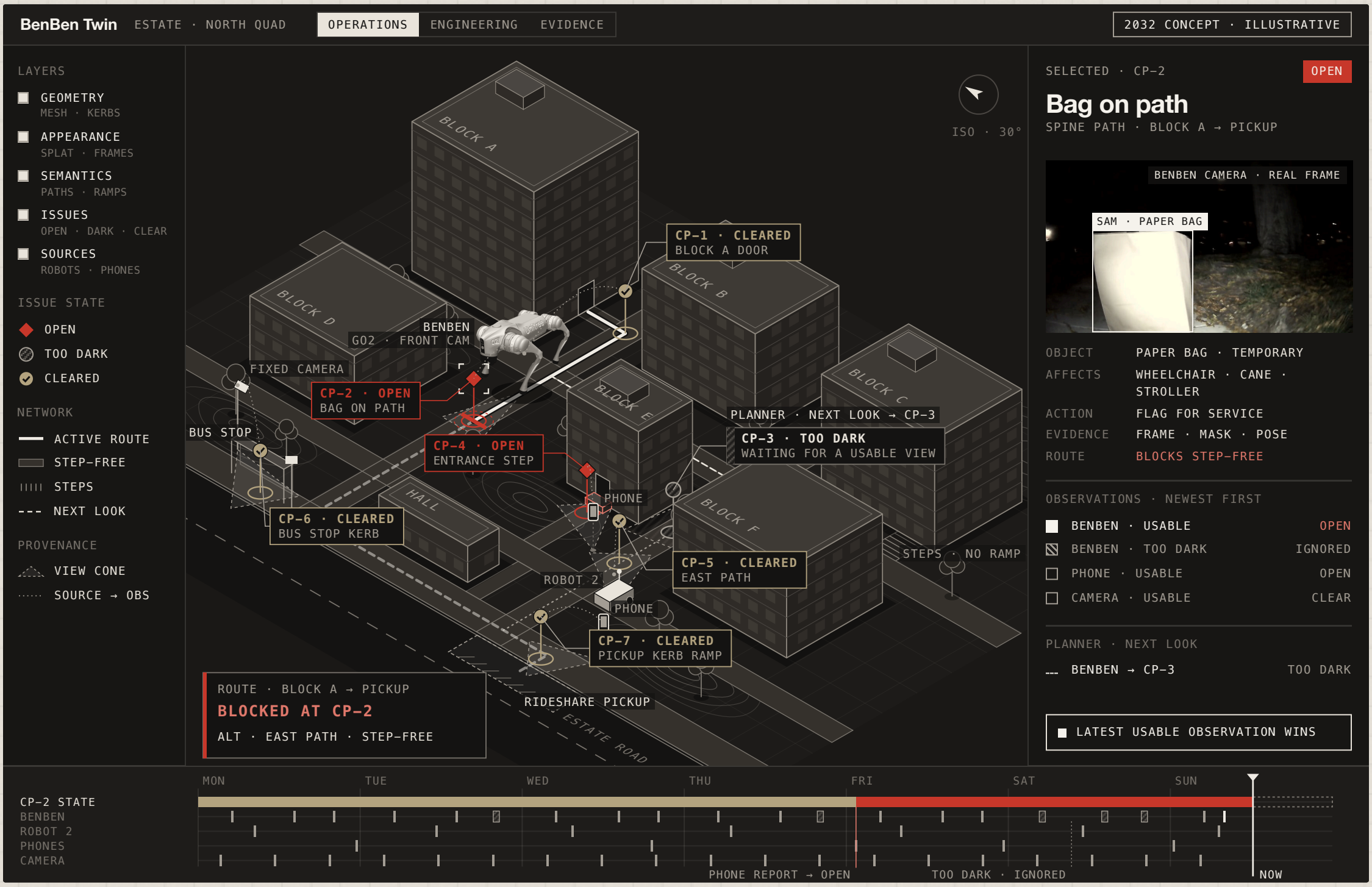
Every robot, phone and camera feeds one memory of a place. **Nothing is called fixed until it’s seen fixed.**

01 THE TWIN · ONE ROUTE, EVERY SOURCE



02 PERCEPTION CASCADE · PLACE MEMORY

TIER 0 — FREE

Is this view usable? Did the place change?

IMAGE QUALITY · POSE · LIDAR GEOMETRY CHANGE

TIER 1 — LOCAL

What changed?

META OPEN WEIGHTS ON THE DEVICE · SAM 2.1 · V-JEPA 2*

TIER 2 — API

Is it a barrier? What’s the evidence?

MUSE SPARK + SAM 3.1 · JUDGE AND CITE

ONE CHECKPOINT ACROSS VISITS

VISIT 1 · ROBOT	VISIT 2 · PHONE	VISIT 3 · ROBOT	VISIT 4 · CAMERA
Clear	**Blocked**	**Too dark**	**Cleared**
USABLE · CLEAR	USABLE · **BLOCKED**	NOT USABLE · **STAYS BLOCKED**	USABLE · CLEARED

Latest usable observation wins. A dark frame or an empty detection never clears an issue.

03 HOW IT SCALES

AUTONOMY

Patrol the open world, planned live.

- REASONER — Muse picks the next look
  open issues, stale checkpoints, reports
- STATE — The twin and its place memory
- ACTION — A navigation policy drives
  promoted only if it beats replay and sim

MOTION GUARD · HUMAN OVERRIDE · NO TEXT-TO-MOTOR

SCALE

Verified access states, fed to routing apps.

Estates · Campuses · Hospitals

Verified access state
per checkpoint, with its evidence

Routing apps
Emerald Path-style step-free routing

GUARDRAILS

People see the evidence before anything happens.

- Privacy
  Blur people before anything is stored or shared.
- Human review
  A person approves every external action.
- Evidence trail
  Every state links to the frames behind it.

WHAT WE’D MEASURE

Named, not yet measured.

Missed changes · False clears · Citation validity · Time to verified fix · Duplicate actions

Meta stack Muse Spark · SAM 3.1 · MapAnything · Voice Transcribe

github.com/shivraj-S-bhatti/benben